

AI Practitioner Skills Framework

Print edition - v5.1 - August 2026 - 7 core domains - License CC-BY-SA 4.0

01 - Technical Prompt Engineering

SUB-SKILL	EXAMPLE
Scaffold Method	A woman walking in rain + cinematic lighting + 85mm f/1.8 + photorealistic + 4K
Information Priority (Front-Loading)	Oil painting of a castle... (style first) vs. A castle, oil painting...
Photographic Vocabulary	85mm lens, f/2.8, shallow depth of field instead of hyperrealistic
Active Voice Editing	remove the background, add a red hat, turn the car into a truck
Agent Prompting Patterns	You are a cinematographer. First analyze the scene mood, then specify lighting.
Long-Horizon Video Scripts	Scene 1: Close-up on protagonist, golden hour. Transition: Cross-dissolve to wide establishing
Camera Movement Language	pull from close-up to wide shot, circular tracking shot, dolly zoom
Advanced Prompt Syntax	(masterpiece:1.5), (best quality:1.3), [cinematic lighting] :: 0.8
Semantic Layering	A [serious executive] [in shadows] [dramatic side lighting] creates different emphasis than [dr
Style Injection Techniques	::0.3 after style reference to preserve subject while adding aesthetic influence

02 - Photographic Literacy

SUB-SKILL	EXAMPLE
Lighting Pattern Mastery	Rembrandt lighting: key light 45deg high, fill at 1/4 power, subtle rim for separation
Lens Selection	85mm portrait lens for flattering facial compression and background bokeh
Focal Length Mapping	24mm at 2m = dramatic wide-angle distortion; 135mm at 5m = flattering compression with natural
Optical Physics for AI	Prompt: shot on Leica 50mm Summilux, visible anamorphic horizontal flares, organic bokeh circle
Aperture Control	f/2.8 for subject isolation; f/11 for product shots requiring full sharpness
Native Resolution Management	Generate at 4K native to capture skin pores and fabric weave; avoid upscaling artifacts
Vertical Format Mastery	Center subject with negative space above for TikTok text overlays; use leading lines vertically
Advanced Rendering Terms	subsurface scattering on skin/ears; ambient occlusion in fabric folds and crevices
Anamorphic Mastery	Prompt: shot on Panavision Primo anamorphic, horizontal blue lens flares, 2.39:1 aspect ratio,
Cinematic Grammar	Open on wide establishing shot, push in to medium, cut to close-up reaction; parallel editing a
Executive Portraiture	Prompt: CEO portrait, corner lighting establishing authority, dark navy suit against library ba
Quiet Luxury Aesthetic	Prompt: quiet luxury editorial, cream cashmere sweater, matte gold jewelry, no visible logos, s

03 - Strategic Negation & Skin Mastery

SUB-SKILL	EXAMPLE
Negative Prompting with Weights	(plastic skin:1.4), (cartoon:1.3), (extra fingers:1.4)
Skin Realism Management	Prompt: visible pores, fine vellus hair, subtle skin variation Negate: airbrushed, doll-like, w
Anatomical Correction	(fused fingers, extra limbs, double iris, asymmetrical eyes:1.4)
Temporal Consistency for Video	Use --cref with character sheet + retraining by error recycling for stable sequences
Drift Correction	Generate 10-frame clip, identify drift points, retrain prompt weights, regenerate with correcti

04 - Identity Preservation & Consistency

SUB-SKILL	EXAMPLE
Seed Locking	--seed 12345 to reproduce base composition while varying style modifiers or lighting
Reference Tools	--cref character_sheet.jpg --sref art_style.jpg --cw 80
Character Weight (--cw)	--cw 80 to preserve face and clothing but allow pose variation and expression changes
Multi-Reference Consistency	char1.jpg + char2.jpg + background.jpg + prop.jpg for ensemble scene with consistent elements
Agent-Based Consistency	DirectorAgent coordinates Veo 3.1 + Midjourney for cross-scene visual continuity and script fid
Visual-Script Alignment (VSA)	Score generated frames against storyboard; flag deviations >15% for review and regeneration

05 - Post-Processing & Hybrid Workflows

SUB-SKILL	EXAMPLE
Iterative Refinement (Branching)	Keep --seed 12345; change weather: rainy to snowy; compare outputs side-by-side
Inpainting & Outpainting	Fix malformed hands via targeted inpainting; extend background via intelligent outpainting
External AI Enhancement	Upscale 4K native gen to 8K with Topaz; color grade in Lightroom for cinematic LUT application
Unified Generation-Editing	Iterate on image + text overlay + composition adjustments in single interface without context s
Cross-Domain Compositing	Composite illustrated character into photoreal background via precise masked inpainting and lig
Text Overlay Integration	Prompt: text: "LAUNCH 2026" in bold serif, bottom-center, gold foil texture with subtle drop sh
Post-Generation Mastery Checklist	Check hands/feet anatomy, verify eye direction consistency, confirm lighting coherence, test at

SUB-SKILL	EXAMPLE
Model Serving	REST/gRPC endpoints, Kubernetes orchestration, load balancers, auto-scaling groups
Quantization & Optimization	ONNX export, TensorRT, GGUF quantization, pruning, distillation techniques
Monitoring & Logging	Prometheus/Grafana dashboards, custom telemetry, LangSmith, structured logging
CI/CD for AI Pipelines	GitHub Actions, MLflow, model versioning, automated regression testing, canary deployments

06 - AI Agent Orchestration

SUB-SKILL	EXAMPLE
Agent Architecture Design	Agent workflow: search web, summarize findings, generate image, draft social post, schedule pub
Router Pattern Implementation	Query "create video ad for product launch" routed to VideoAgent; "write blog post" routed to Te
SLM Optimization	Use 7B SLM for email drafting and summarization; reserve 70B model for complex strategy and cre
Role-Based Agent Teams	PlannerAgent breaks brief into tasks, ResearcherAgent gathers references, WriterAgent drafts, R
Cross-Agent Communication	Agents publish/subscribe to task_context channel; use consensus voting for creative decisions
Autonomous Research Workflows	ResearchAgent tasked: "Gather Q1 2026 AI video generation trends from 5+ sources, output markdo
Private Skill Deployment	compliance-check.skill validates all outputs against company brand guidelines, legal requiremen
Auditable Agent Workflows	Log: Agent X called tool Y at Z time with input A, output B, human approval C, final deployment
Sovereign AI Infrastructure	Run agent cluster in private cloud; all data stays within organizational boundaries; zero exter
Skills Packaging (SKILL.md)	This document is a SKILL.md example; agents load only relevant sections per task via progressiv

07 - LoRA Ecosystem & Parameter-Efficient Fine-Tuning

SUB-SKILL	EXAMPLE
LoRA Fundamentals	<code>r=16, lora_alpha=32, target_modules=["to_q","to_k","to_v"]</code>
QLoRA Optimization	Fine-tune 7B model on single 12GB GPU with 4-bit quantization + LoRA rank 16
DoRA (Weight-Decomposed)	<code>decompose=True</code> in LoRA config for detail-sensitive portrait style adaptation
AdaLoRA (Adaptive Rank)	<code>rank_budget=0.5</code> - auto-allocates higher rank to critical attention layers
Rank-Stabilized LoRA (rsLoRA)	<code>rscale=True</code> - enables rank 64+ without training instability
LoRA+ (Differential LR)	<code>lr_A=1e-4, lr_B=1e-3</code> - B matrix learns 10x faster for better optimization
VeRA (Vector-based)	Train only 0.01% of parameters using shared frozen random bases
GLoRA (Global Prompt)	<code>task_descriptor -> global_adapter -> per_layer_lora</code> for style switching
LoRA Style Libraries	<code>style_noir@0.6 + lighting_rembrandt@0.4 + subject_ceo@0.8</code>
LoRA Training Pipeline	20-50 curated images -> QLoRA rank 16 -> 10-20 epochs -> safetensors export
S-LoRA Serving	Load 50+ LoRA adapters dynamically per request with KV cache sharing
LoRA x Framework Integration	Character LoRA x style LoRA x lighting LoRA combined for brand campaign consistency

Platform & Tool Reference

Pro Image

Tools: Flux, Ideogram, Recraft, Qwen Image 2.0

Video Generation

Tools: Veo 3.1, Wan 2.6, Sora, Runway Gen-3

Hybrid Editing

Tools: Photoshop (Gen Fill), ComfyUI, Topaz Photo AI, ACE++

Agent Frameworks

Tools: LangChain, AutoGen, CrewAI, custom builders, SKILL.md parsers

Deployment

Tools: vLLM, TensorRT, ONNX Runtime, GGUF, Kubernetes

Monitoring

Tools: Prometheus/Grafana, LangSmith, Arize, custom telemetry

Resources

Interactive site

<https://marktantongco.github.io/aiskills-photog/>

Print wiki (HTML)

<https://marktantongco.github.io/aiskills-photog/wiki.html>

Source document

<https://marktantongco.github.io/aiskills-photog/SKILL.md>